NOT MEASUREMENT SENSITIVE
MIL-P-85856/24
7 December 1988

MILITARY SPECIFICATION SHEET

POWDER MATERIALS FOR FLAME AND PLASMA SPRAYED COATINGS TUNGSTEN CARBIDE-COBALT, TYPE I, CLASS 1, COMPOSITION AA

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the powder described herein shall consist of this document and the latest issue of MIL-P-85856.

Chemical Composition

Fe	W	C	Co	Others	
				Each	Total
2.0 max.	Bal	4.8 5.6	15.0 18.0	-	2.0 max.

Particle Size/Distribution

+270 Mesh (+53 µm)	15.0 max.
-270 Mesh (-53 µm)	85.0 min.
(-20 µm	15.0 max.

Metallographic Inspection

Powder particles shall be capable of exhibiting a clustered appearance consisting of tungsten carbide-cobalt aggregates in contact with itself and cobalt.

Sprayed Coating Properties

Bond Strength	62.1 MPa (9000 psi) min.
Hardness	800 to 1000 VHN 300.
Microstructure	See paragraph 4.5.4.1.2 Porosity - 3% max. Unmelted Particles - 3% max.

AMSC N/A
FSC 3439
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Deposit Efficiency and Deposition Rate

60% min for A/H_2 @ 11.0 lb/hr

Comparable Documents

Powder which meets the requirements specified is qualified under the following specifications.

General Electric Company	B50TF167, Class A
Rolls-Royce Ltd.	MSRR 9507/1

Custodians:
Air Force - 20
Navy - AS
Army - MR

Review activities:
Air Force - 99, 71
Navy - SH

Preparing activity:
Navy - AS
(Project No. 3439-0663/24)

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